07 - Machine Learning and Fraud Detection

Models, training, tuning, anomaly detection, evaluation

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

src/fraud_detection/train.py; src/fraud_detection/risk.py; src/anomaly_detection/detectors.py; src/model_evaluation/evaluate.py; models/model_metadata/last_run_summary.json

Algorithms used and why

Model	Type	Simple explanation	Imbalance handling	Tuned parameters (search space)
Logistic Regression	supervised	Weighted sum of features turned into a probability; easy to explain to auditors	class_weight=balanced	C (log-uniform 1e-3..1e2), after StandardScaler
Random Forest	supervised	Many decision trees vote; handles non-linear patterns	balanced_subsample	max_depth 4-15, min_samples_leaf 2-29, max_features sqrt/0.3/0.5
XGBoost	supervised	Trees added one by one, each correcting earlier mistakes	scale_pos_weight = legit/fraud ratio	max_depth 3-7, learning_rate 0.02-0.3, subsample, colsample_bytree, min_child_weight
Isolation Forest	unsupervised	Random splits isolate odd points quickly; short path = anomaly	contamination	none; fitted without labels
Local Outlier Factor	unsupervised	Compares a point's local density with its neighbours	contamination	n_neighbors=35 on a 6000-row sample

Training procedure (src/fraud_detection/train.py)

- Chronological split; features are the 40 columns from feature_names().
- For each enabled model, RandomizedSearchCV (stratified k-fold on the training set, score = average precision) picks hyper-parameters and refits.
- The decision threshold is chosen on the validation set to maximise F1 (src/model_evaluation/evaluate.py: best_f1_threshold).
- The test set is scored once with the saved threshold. The champion is the supervised model with the highest PR-AUC.
- Models are saved with joblib and a JSON metadata file (parameters, CV score, threshold, features, metrics, data hash, seed, library versions).

Risk classification

Probability	Tier	Action
< 0.05	low	allow
0.05 - 0.3	medium	step-up authentication
0.3 - 0.7	high	hold for manual review
>= 0.7	critical	block + alert

A financial example

Suppose 10,000 card payments contain 200 frauds. A model that flags nothing is 98% accurate yet catches 0 frauds (recall 0). A model that flags 300 payments, 150 truly fraudulent, has precision 0.50 and recall 0.75; its 150 wrong flags are legitimate customers whose card was challenged. The false-positive rate is 150/9,800 = about 1.5%. This illustrative arithmetic is only an example of the metrics; it is not a result of this project.

Actual results from the last run

Run run_20261109_061947_57b9, seed 42. Test-set metrics at validation-tuned thresholds. All data are synthetic.

Model	Type	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR
logistic_regression	supervised	0.9901	0.883	0.563	0.688	0.872	0.666	0.0015
random_forest	supervised	0.9925	0.928	0.667	0.776	0.903	0.741	0.0010
xgboost	supervised	0.9925	0.982	0.626	0.765	0.885	0.743	0.0002
isolation_forest	unsupervised	0.9743	0.365	0.437	0.398	0.780	0.400	0.0150
local_outlier_factor	unsupervised	0.9618	0.187	0.287	0.226	0.683	0.116	0.0248

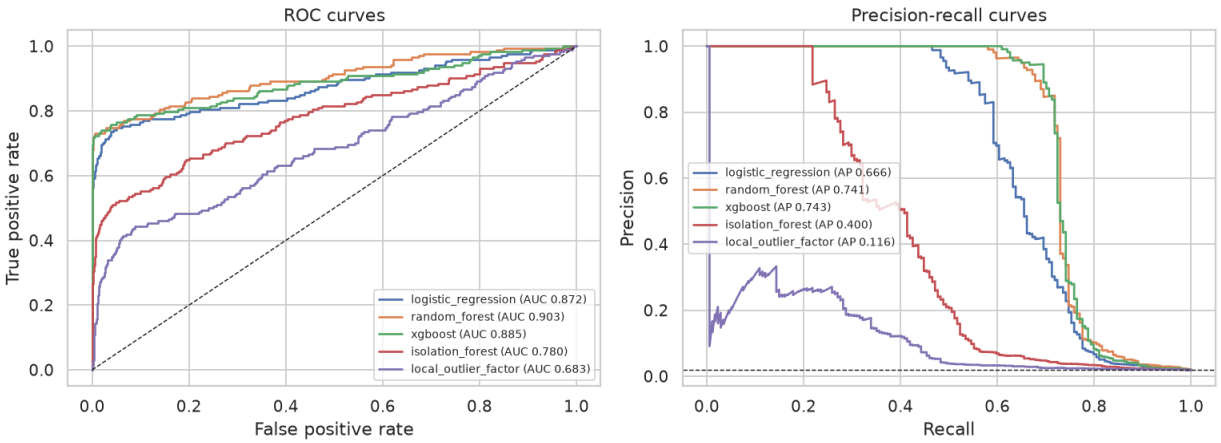
Model	TP	FP	FN	TN	Threshold
logistic_regression	98	13	76	8766	0.929
random_forest	116	9	58	8770	0.671
xgboost	109	2	65	8777	0.902
isolation_forest	76	132	98	8647	0.530
local_outlier_factor	50	218	124	8561	1.380

Model	CV average precision	Train+tune seconds	Best parameters
logistic_regression	0.6214	28.1	{'clf__C': 0.0019517224641449498}
random_forest	0.7226	164.6	{'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 16}
xgboost	0.7251	26.6	{'colsample_bytree': 0.6624074561769746, 'learning_rate': 0.03051362379613698, 'max_depth': 5, 'min_child_weight': 8, 'subsample': 0.7334834444556088}

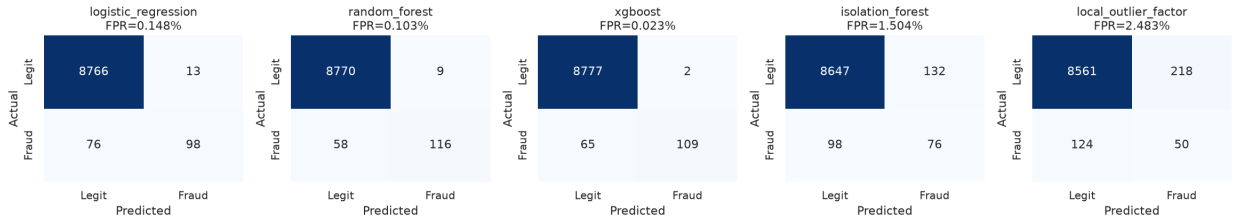
Champion: xgboost. Bootstrap 95% CI of its ROC-AUC: [0.847, 0.918]. McNemar vs Logistic Regression: {'b': 30, 'c': 8, 'chi2': 11.605263157894736, 'p_value': 0.0006576543953178447}.

Risk tiers on the test set

Tier	Transactions	Frauds	Observed fraud rate	Action
low	3	0	0.0000	allow
medium	7871	37	0.0047	step-up authentication
high	912	12	0.0132	hold for manual review
critical	167	125	0.7485	block + alert



ROC and precision-recall curves (generated by the pipeline).



Confusion matrices at validation-tuned thresholds (generated by the pipeline).

Security-event anomaly detection

Isolation Forest fitted without labels on the first 70% of time: ROC-AUC against known attacks 0.994; 3550 events flagged; attack block rate 0.797.

Implemented vs not implemented

Item	Status
LR, RF, XGBoost with tuning	IMPLEMENTED
Isolation Forest, LOF	IMPLEMENTED
Bootstrap CI, McNemar test	IMPLEMENTED
Probability calibration step in pipeline	MISSING
SHAP / explainability	PLANNED
Multi-seed repeated experiments	PLANNED
Concept-drift handling	PLANNED
Public ULB benchmark results	PENDING